

TrackLift-Pro: Technical Manual

1. Introduction and Software Architecture

TrackLift-Pro is a cross-platform desktop application built using the **Electron** framework by vite coding with Antigravity IDE. The purpose of this software is to assist in railway maintenance by optimizing the vertical track profile for improving the running. By packaging Node.js backend capabilities alongside a Chromium frontend, the application provides a seamless desktop experience without the need for complex server deployments.

The user interface is constructed using standard web technologies: HTML5, CSS3, and JavaScript (ES6+). Styling is accomplished using vanilla CSS, ensuring a highly responsive, performant, and modern aesthetic without relying on heavy external framework dependencies. All data processing and intensive mathematical algorithms are executed locally on the client side via the V8 JavaScript engine, resulting in fast, native-like performance even when processing thousands of track stations.

2. Mathematical Foundations & Algorithms

The core engine of TrackLift-Pro relies on advanced heuristic optimization algorithms to compute the smoothest possible vertical track profile while respecting real-world lift and lower constraints. The software tries to bring the proposed levels within the permitted lift/lower and the unevenness/SD limits. After the optimization, the software mathematically drapes a smooth **Cubic Spline** over those points (Proposed rail levels) to export the continuous curved proposed rail profile.

2.1 Unevenness(V) Calculations

Unevenness or vertical versine is defined as the offset of the track from a theoretical chord connecting two points. In order to calculate the unevenness, short chord of 20m length and long chord of 80m length is used. The 20m/80m chords against 9m/18m chords in TRC will give much better analysis about the track condition. The default limits for unevenness for short and long chord are 5mm and 40mm respectively. These values are much strict than the New Track Tolerances as per IRPWM 2024 para 520.

2.2 Standard Deviation (SD) Calculations

Standard deviation represents the un-evenness of the track over a running window. In this software we adopted 200m rolling block for calculating the SD. i.e. for calculating the SD for a particular station, the unevenness values of 100m behind the station and 100m ahead the station is used. This continuously rolling block, unlike the fixed block for SD calculation in TRC, will give much realistic values. The default limits for SD values for short and long chord are 2mm and 20mm respectively.

In short, both the default Unevenness/SD value limits for short/long chords are much strict than the New Track Tolerances as per IRPWM 2024 para 520. Hence the proposed rail profile will offer a satisfactory riding experience. However, the user is having freedom to relax these limits as per site conditions.

2.3 Optimization Engine (Simulated Annealing)

The solver employs a heuristic optimization algorithm inspired by **Simulated Annealing**. This is a probabilistic optimization algorithm used to find the global maximum or minimum of a complex function over a large search space. It mimics the metallurgical process of heating and slowly cooling a material to minimize structural defects and achieve a stable state of minimum energy.

In the context of TrackLift-Pro, the 'cost function' is a mathematical penalty score representing how 'bad' the track geometry is (high versine, high standard deviation, or excessive lift/lowering). The goal of the engine is to find the track profile with the lowest possible cost. The detailed workflow is as follows:

Phase 1: Initialization

- The system starts by copying the 'Existing Level' as the initial 'Proposed Level'.
- A virtual 'Temperature' is initialized at a high value. This temperature dictates how much 'chaos' or randomness is allowed in the system.
- The initial Global Cost is calculated based on versine limit violations, SD violations, and lift/lower penalties.

Phase 2: Iterative Randomization (The Annealing Loop)

For each iteration (up to the user-specified limit, e.g., 100,000):

- The engine selects a random, non-locked station and applies a tiny, random adjustment (up or down) to its Proposed Level, constrained by the Custom Max Lift/Lower limits.
- The software recalculates the entire track's Global Cost with this new adjustment.

Phase 3: Acceptance Probability

- If the new cost is LOWER (the track got smoother), the adjustment is immediately ACCEPTED.
- If the new cost is HIGHER (the track got worse), the adjustment is NOT immediately rejected. Instead, the engine uses the current 'Temperature' to calculate an acceptance probability. When the temperature is high (early in the process), it frequently accepts 'worse' profiles. This is a critical mathematical trick to allow the algorithm to escape 'local minima' (getting trapped in a sub-optimal but decent shape) and explore a wider range of possibilities to find the true 'global minimum' (the absolute smoothest profile).

Phase 4: Cooling Schedule

- As iterations progress, the virtual 'Temperature' slowly cools down (decreases).
- As it cools, the probability of accepting 'worse' profiles approaches zero. The engine transitions from wild exploration into precise, localized fine-tuning.

Phase 5: Finalization

- Once the temperature reaches zero or the max iterations are hit, the final Proposed Levels are locked in, delivering the mathematically smoothest track profile possible under the given lifting constraints.

2.4 Lift vs. Lower Bias Weightage

In railway maintenance, lifting the track with ballast is structurally sound and standard practice. Lowering the track requires removing core ballast, which is a tedious job and generally resorted during regular maintenance. In view of the above, the software is designed in such a way that, the solver explicitly hardcodes a 10x penalty multiplier to any 'lowering' adjustments. This mathematically biases the Projected Gradient Descent to overwhelmingly favour lifting the track. That is, even the max lower limits are permitted, system will look for achieving the result with lifting (within max lift range) at 10:1 lift to lower ratio. This will help in obtaining results with minimum lowering possible.

2.5 The Hybrid Engine: Cubic Spline & Linear Interpolation

In order to make this software handy for field units during execution, an additional tamping PDF report can be generated after optimization, which will give lift/lower values at every 3.33m intervals. Generating intermediate stations will not affect the optimized results. It will simply interpolate the levels and lift/lowering values at shorter intervals. TrackLift-Pro utilizes a unique Hybrid Engine:

- Proposed Profile (Cubic Spline): The optimized track levels are perfectly smooth. A Natural Cubic Spline algorithm is applied to mathematically trace the physical elastic bending of the rail curve, outputting micro-precise 3.33m intermediate marks without breaking the curve's fluidity. This is far superior to linear interpolations done at field level and give much satisfactory results.
- Existing Profile (Linear Interpolation): The existing (un-tamped) track geometry is jagged and bumpy. If a cubic spline were applied here, it would create artificial 'overshoots' and phantom potholes in the mathematical model. To prevent this, stable Linear Interpolation is actively used to suppress these artificial spikes, yielding accurate existing levels.

3. User Interface Guide (Buttons, Panels, and Cells)

This section details the purpose of every interactive element in the software.

3.1 Project Details Panel

- Zone / Division / PWAY Section: Text input fields used to record the metadata of the railway project.
- Km From / Km To: Used to indicate the start and end of the block section being evaluated.
- These details are automatically printed on the headers of all exported PDF and Excel reports.
- All the entries in this panel are optional.

3.2 Configuration & Constraint Panel

This panel contains the absolute boundaries and limits that guide the optimization engine. Any proposed track profile must strive to exist within these parameters:

- Start Chainage, End chainage: Starting/Ending Km of survey/execution. Can be entered as 500.000,500.0000,500/000,500/0000 formats.
- Interval (m): Defines the mathematical distance between surveyed stations. **The Rule for Intervals:** For the 20m and 80m chord checks to be valid, your station interval MUST be a clean divisor of 10m. The software is explicitly designed to handle standard **10m** intervals (or smaller, like **5m** or **2m** or **1m**). (Note: While importing excel data, system will calculate the distance between first 2 stations and assume the same interval for all other stations.)
- Default Max Lift (mm): Establishes the global hard ceiling for raising the track. In railway engineering, lifting the track is structurally sound and preferred. This sets the maximum allowable upward movement for any station unless overridden by a Custom Max Lift cell.
- Default Max Lower (mm): Establishes the global hard floor for dropping the track. Since lowering the track is a tedious job, the software is designed to penalize lowering at a rate of 10x when compared to lifting. It is preferable to keep this value as 0 and perform optimization, then gradually increase it until the desired output is achieved.
- Target V20 Limit (mm): The maximum allowable deviation for the 20m short chord. Restricting this limit forces the engine to eliminate sharp, high-frequency track defects that cause intense vertical train vibrations.
- Target V80 Limit (mm): The maximum allowable deviation for the 80m long chord. Restricting this limit prevents long-wave geometry sweeping that could cause train suspension resonance at high speeds.
- Target SD20 / SD80 Limits: Standard Deviation limits dictate the overall permitted 'roughness' over a sliding window of 200m. Even if individual versine pass, if the track is consistently jagged, the SD will fail.
- Generate grid: Generates a grid to enter existing track details based on the start chainage, end chainage and interval values.
- Template: Download an excel template file in which data can be imported.
- Import Excel: Data can be imported as excel files in the format of template file.
- Reset: Hard reset all the data already entered. All the unsaved data/result/report will be deleted. Use with caution.

3.3 Data Entry Panel

- Search Chainage: Searching the data for a particular chainage.
- Go to Optimization Panel: Directly scrolls down to Optimization panel bypassing data entry panel.
- Lock Checkbox: By ticking the 'Locked' box on a station, the user forces the optimizer to maintain the exact existing level at that point (Lift/Lower becomes exactly 0). First and last stations are always auto locked to avoid lifting at start/end stations.
- No, Chainage: Auto generated based on the start chainage, end chainage and interval values.
- Existing level: Enter the existing rail levels against the autogenerated chainages. Can be entered as 500.000,500.0000,500/000,500/0000 formats.
- Proposed Level: If you keep the column blank and run optimizer, the system will treat the existing levels as proposed for initializing the optimization. If separate values are given, the software will load the provided proposed levels and *immediately* calculate the resulting Lift/Lower values, the 20m and 80m versines, and the block-wise Standard Deviations. If you *do* decide to click "Start Optimization", the engine will use your imported proposed profile as its starting foundation. It will begin mathematically smoothing the track from *your* design, rather than starting from scratch, saving a massive amount of computational time.
(Pro-tip: If you want to force the software to keep your proposed levels exactly as they are without the optimizer ever touching them, you can lock your rows)
- Max lift/Max lower: Station wise lift/lower values can be given, overriding global values. If left blank, global values will be considered. Useful where lifting/lowering is to be limited for few stations.
- Remarks: Any custom data regarding the station can be added. Same will be shown in the reports generated.

3.4 Alignment Optimization Engine

- Smoothness Emphasis Factor: The Smoothness Emphasis Factor is the primary control knob for tuning the aggressiveness of the optimization engine. In the backend cost function, the software balances two competing desires: (1) Minimize structural track movement (stay close to the existing track), and (2) Minimize versine/SD deviations (make the track perfectly flat). This dropdown dictates which desire wins. The software offers four levels of emphasis:

- **Low (Weight = 1):** The solver prioritizes minimizing the sheer volume of track lifting. It will allow small humps and dips to remain in the profile, hugging the existing jagged track tightly, provided the deviations barely pass the safety limits. Ideal for areas where lifting is extremely difficult.
- **Medium (Weight = 10):** Standard track alignment smoothing. Provides a balanced compromise between achieving a comfortable ride quality and conserving ballast usage.
- **High (Weight = 50):** The solver becomes highly sensitive to track roughness. It aggressively flattens humps and dips by pushing the track levels up to the maximum allowable Custom Lift limits.
- **Extreme (Weight = 200):** The solver applies an overwhelming mathematical penalty to any microscopic geometric defect. This forces the engine to ignore ballast conservation completely, relentlessly lifting the track to achieve a perfectly flat, glass-like running surface. Ideal for high-speed corridors.

These numbers (1, 10, 50, and 200) are the exact **mathematical multipliers** used in the software's backend equation.

During optimization, the software tries to achieve the lowest possible "Penalty Score." It gets penalized for two things:

1. **Lifting the track** (which costs ballast). Let's say lifting the track by 1mm gives a penalty of **1 point**.
2. **Leaving a geometric defect** (a versine bump or dip). Let's say leaving a 1mm defect gives a penalty of **1 point**.

The "Weight" is a multiplier applied *only* to the defect penalty:

- ✓ **Weight = 1 (Low):** A 1mm track defect adds 1 point to the penalty score. Lifting the track by 1mm also adds 1 point. The software treats them as equally bad. It will only lift the track if it absolutely has to, saving you ballast but leaving the track slightly bumpy.
- ✓ **Weight = 10 (Medium):** A 1mm defect now adds **10 points** to the penalty score. Now, the software realizes that leaving a bump is 10 times worse than using up ballast. It will happily lift the track by up to 9mm just to fix a 1mm bump.
- ✓ **Weight = 200 (Extreme):** A 1mm defect now adds a massive **200 points** to the penalty score. The software becomes mathematically "terrified" of leaving even a microscopic deviation in the track. Because the penalty for a defect is so overwhelmingly high, the software will ignore ballast conservation and lift the track up to the maximum permitted limits to achieve a mathematically smooth surface.
- **Max Iterations:** Controls the computational depth of the Simulated Annealing engine. For example, '100000' instructs the solver to evaluate 100,000 distinct, randomized track combinations to find the absolute lowest global cost. Higher values guarantee smoother tracks at the expense of slightly longer processing time
- **Run Auto Smoother:** Locks in all configurations and triggers the backend solver. The duration of optimizer depends upon the number of stations and Max iterations.

3.5 Interactive Graph Features

The software provides two visual dashboards powered by dynamic charting algorithms, designed to provide a macro-level overview of the track geometry:

- **Profile vs. Chainage Graph:** Overlays the rugged Existing Profile (red/blue) with the smooth Proposed Profile (green). This visualizes how the optimization engine has mathematically 'ironed out' the track.
- **Lift/Lower vs. Chainage Graph:** A bar chart that explicitly graphs the physical machine operations required at each station. Spikes indicate areas requiring heavy tamping.
- **Hover Data Tooltips:** Users can hover their cursor over any point on the graphs to instantly view the exact Chainage, Existing Level, Proposed Level, and calculated Lift/Lower amount without digging through the data table. Similarly, when hovered over bar chart, it gives additional details regarding that station.
- **Horizontal Pan & Zoom Slider:** For large datasets (e.g., several kilometres of track), the graphs can become visually dense. The slider bar at the bottom allows users to dynamically zoom into specific blocks and pan horizontally to visually inspect specific track defects up close.

3.6 Optimization & Compliance Dashboard

- Shows the status of track unevenness w.r.t the constraints given by user. A compliance status "Pass" means proposed profile achieved within the prescribed limits.

3.7 Result Table & Display Cells

- **Existing vs Proposed Level:** Displays the original surveyor level vs the newly calculated optimal level (in meters).
- **Lift/Lower (mm):** Displays the amount of lifting/lowering required at that station for achieving the proposed levels. Positive value indicate lifting and negative value indicates lowering. (Note: General lift, if any, for tamping machines should be given separately)

- Results for 20m/80m chord for both existing track and Proposed track. The side-by-side comparison of each parameter for existing vs proposed track gives the exact improvement to track after optimization. Any values outside limits will be shown as red. It should be noted that after optimization, the status for passing all criteria purely depend up on the constraints given and existing condition of the track. For a highly disturbed track, higher lifting/lower values may be required to pass all criteria. Otherwise, user may give relaxed values for unevenness and SD. In any case, the track parameters will be improved from existing track condition.

3.8 Export Buttons

- Export PDF Report: Generates a professionally structured PDF containing the full details in the result table at 10m intervals.
- Export Tamping PDF: Specifically designed for tamping machine operations. Calculates and prints two intermediate stations (at 3.33m and 6.67m) between every 10m standard station using the Hybrid Spline Engine. This report shows only Chainage, Final level, lift/lowering and Remarks.
- Export Excel: Exports all results to an '.xlsx' spreadsheet containing the full details in the result table at 10m intervals.

4. Technical Constraints & Limitations

4.1 Virtual Boundary Extrapolation

To compute the unevenness with 20m chord, 1 station behind and 1 station ahead is required. However, this condition is not met for first and last station (similarly, the first 4 and last 4 stations for an 80m chord). At these stations, the software utilizes linear extrapolation. It projects the existing and proposed gradients out into 'virtual space' for computation purpose. However, these virtual calculations are hidden from the user interface to prevent analytical confusion during inspections. So, it is desirable to start/end the survey with locations having minimum disturbances or sudden grade change for at least 50m, for getting most accurate results.

4.2 Fixed SD Block Exception (First & Last 100m)

Standard Deviation (SD) is calculated over a 200m "rolling window" mathematically centred on the current station. This means the engine tries to look exactly 100m backwards and 100m forwards. However, for the stations located at the physical beginning or end of your dataset, there simply isn't enough surveyed track data available to look 100m backwards or forwards.

To solve this, the software implements a Fixed Block Exception:

First 100m: For any station located within the first 100 meters, the software cannot roll backwards. Instead, it "locks" the window to evaluate the absolute first 200m chunk of the track. The reported SD for all these early stations will be identical, representing the overall roughness of that specific 200m entrance block.

Last 100m: Similarly, the software cannot roll forwards past the end of the dataset. It locks the window to evaluate the absolute final 200m chunk of the track.

Middle Section: For every single station between the first and last 100m, the software has enough data on both sides and reverts to the standard dynamic, rolling window perfectly centred on each station.

5. Conclusion

TrackLift-Pro seamlessly integrates powerful node-based computational algorithms with an intuitive front-end architecture, delivering unparalleled precision in vertical track profiling. The integration of Natural Cubic Splines for interpolation and Simulated Annealing for geometry optimization ensures an industry-standard output ready for immediate deployment for tamping operations.

6. Feedback

All efforts are taken to align the software with the field working conditions. Any suggestions, corrections required, bugs identified may be communicated to shanilgd@gmail.com. I will try to improve those deficiencies in future OTA updates.